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Doherty amplifier

Abstract

A Doherty amplifier includes a first amplifier that includes first output fingers and a first output electrode connected to the first output fingers, a second amplifier that includes second output fingers and a second output electrode connected to the second output fingers, a first bonding wire connected between a first region in the first output electrode and a second region in the second output electrode, a second bonding wire connected between a third region in the first output electrode and a fourth region in the second output electrode, and at least one of a first capacitor connected in series with the first bonding wire, and a second capacitor connected in parallel with the second bonding wire, wherein the first and the third regions are regions to which the first output fingers are connected, and the second and the fourth regions are regions to which second output fingers are connected.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8228123	12/2011	Blednov	N/A	N/A
2011/0204980	12/2010	Blednov	330/295	H03F 3/195
2015/0145601	12/2014	Moronval	330/295	H03F 1/07
2021/0075374	12/2020	Wang	N/A	H03F 3/604
2022/0182472	12/2021	Tomita	N/A	H03F 1/07

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2018142178	12/2017	WO	H01L 23/66

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority based on Japanese Patent Application No. 2021-102587, filed on Jun. 21, 2021, and the entire contents of the Japanese patent applications are incorporated herein by reference.

FIELD

(2) The present disclosure relates to a Doherty amplifier.

BACKGROUND

(3) There has been known a Doherty amplifier as an amplifier that amplifies high-frequency signals such as microwaves. In the Doherty amplifier, a main amplifier and a peak amplifier amplify input signals in parallel, and the amplified signals is synthesized by a synthesizer. In the synthesizer, a quarter wavelength line is provided between the main amplifier and a synthesis point. It is known to use an output capacitance and a bonding wire in the main amplifier instead of the quarter wavelength line (for example, Patent Document 1: U.S. Pat. No. 8,228,123).

SUMMARY

(4) A Doherty amplifier according to the present disclosure includes: a first amplifier that includes a plurality of first output fingers and a first output electrode connected to the plurality of first output fingers, amplifies one of two signals into which an input signal is distributed, and outputs an amplified signal to the first output electrode; a second amplifier that includes a plurality of second output fingers and a second output electrode connected to the plurality of second output fingers, amplifies another one of the two signals, and outputs an amplified signal to the second output electrode; a first bonding wire connected between a first region in the first output electrode and a second region in the second output electrode; a second bonding wire connected between a third region in the first output electrode closer to the second output electrode than the first region and a fourth region in the second output electrode closer to the first output electrode than the second region; and at least one of a first capacitor connected in series with the first bonding wire between the first region and the second region, and a second capacitor connected in parallel with the second bonding wire between the third region and the fourth region; wherein the first region and the third region are regions to which the plurality of first output fingers are connected, the second region and the fourth region are regions to which the plurality of second output fingers are connected, and the first amplifier and the second amplifier are arranged in a direction intersecting extension directions of the first output fingers and the second output fingers, respectively.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a block diagram of a Doherty amplifier according to a first embodiment.
(2) FIG. **2** is a circuit diagram of the Doherty amplifier according to the first embodiment.
(3) FIG. **3** is a plan view of the Doherty amplifier according to the first embodiment.
(4) FIG. **4** is a cross-sectional view of a capacitor in the first embodiment.
(5) FIG. **5** is a plan view of a Doherty amplifier in a first comparative example.
(6) FIG. **6** is a plan view of a Doherty amplifier in a second comparative example.
(7) FIG. **7** is a circuit diagram of a Doherty amplifier in a third comparative example.
(8) FIG. **8** is a plan view of the Doherty amplifier in the third comparative example.
(9) FIG. **9** is a plan view of a Doherty amplifier in a first variation of the first embodiment.
(10) FIG. **10** is a plan view of a Doherty amplifier in a second variation of the first embodiment.
(11) FIG. **11** is a plan view of a Doherty amplifier in a second embodiment.
(12) FIG. **12** is a plan view of a Doherty amplifier in a first variation of the second embodiment.
(13) FIG. **13** is a plan view of a Doherty amplifier in a second variation of the second embodiment.
(14) FIG. **14** is a circuit diagram of a Doherty amplifier in a third embodiment.
(15) FIG. **15** is a plan view of the Doherty amplifier in the third embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

(16) The synthesizer may be formed by connecting an output electrode of the main amplifier to an output electrode of the peak amplifier using the bonding wire. However, when the output electrodes of the main amplifier and peak amplifier are wide, a plurality of bonding wires are connected between the output electrodes. In this case, electrical lengths between the main amplifier and the synthesis point via the plurality of bonding wires may differ from each other. This may deviate from the conditions of the ideal synthesizer and degrade the characteristics.
(17) It is an object of the present disclosure to provide a Doherty amplifier that can improve characteristics.

DESCRIPTION OF EMBODIMENTS OF THE PRESENT DISCLOSURE

(18) First, the contents of the embodiments of this disclosure are listed and explained.
(19) (1) A Doherty amplifier according to the present disclosure includes: a first amplifier that

includes a plurality of first output fingers and a first output electrode connected to the plurality of first output fingers, amplifies one of two signals into which an input signal is distributed, and outputs an amplified signal to the first output electrode; a second amplifier that includes a plurality of second output fingers and a second output electrode connected to the plurality of second output fingers, amplifies another one of the two signals, and outputs an amplified signal to the second output electrode; a first bonding wire connected between a first region in the first output electrode and a second region in the second output electrode; a second bonding wire connected between a third region in the first output electrode closer to the second output electrode than the first region and a fourth region in the second output electrode closer to the first output electrode than the second region; and at least one of a first capacitor connected in series with the first bonding wire between the first region and the second region, and a second capacitor connected in parallel with the second bonding wire between the third region and the fourth region; wherein the first region and the third region are regions to which the plurality of first output fingers are connected, the second region and the fourth region are regions to which the plurality of second output fingers are connected, and the first amplifier and the second amplifier are arranged in a direction intersecting extension directions of the first output fingers and the second output fingers, respectively. This makes it possible to improve the characteristics of the Doherty amplifier.
(20) (2) The first output electrode may be separated between the first region and the third region, and the second output electrode may be separated between the second region and the fourth region.
(21) (3) The first bonding wire may not intersect with the second bonding wire.
(22) (4) The first region and the second region may be connected to each other without through the third region and the fourth region, and the third region and the fourth region may be connected to each other without through the first region and the second region.
(23) (5) A signal output by the first amplifier and a signal output by the second amplifier may be synthesized in the second output electrode.
(24) (6) The Doherty amplifier further may include a harmonic processing circuit that is connected to the second output electrode and processes a harmonic component of the signals amplified by the first amplifier and the second amplifier.
(25) (7) A capacitor included in the Doherty amplifier may be the first capacitor.
(26) (8) A capacitor included in the Doherty amplifier may be the second capacitor.
(27) (9) Capacitors included in the Doherty amplifier may be the first capacitor and the second capacitor.
Details of Embodiments of the Present Disclosure
(28) Specific examples of a Doherty amplifier in accordance with embodiments of the present disclosure are described below with reference to the drawings. The present disclosure is not limited to these examples, but is indicated by the claims, which are intended to include all modifications within the meaning and scope of the claims.
First Embodiment
(29) FIG. **1** is a block diagram of a Doherty amplifier according to a first embodiment. As illustrated in FIG. **1**, in the Doherty amplifier, a main amplifier **10** and a peak amplifier **11** are connected in parallel between an input terminal Tin and an output terminal Tout. A high frequency signal is input to the input terminal Tin as an input signal. A distributor **16** distributes the input signal into two signals. One of the distributed signals is input to the main amplifier **10**. The main amplifier **10** (first amplifier) amplifies the input one of the distributed signals and outputs it to a synthesizer **14**. The other signal distributed by the distributor **16** is input to the peak amplifier **11** via a quarter wavelength line **15**. The synthesizer **14** synthesizes an output signal from the main amplifier **10** and an output signal from the peak amplifier **11**, and outputs a synthesized signal to the output terminal Tout.
(30) The main amplifier **10** and the peak amplifier **11** include, for example, FETs (Field Effect Transistors) **12** and **13**, respectively. In the FETs **12** and **13**, a source S is grounded, a high

frequency signal is input to a gate G, and a signal is output from a drain D. The FETs **12** and **13** are, for example, a GaN FET or an LDMOS (Laterally Diffused Metal Oxide Semiconductor). The main amplifier **10** and the peak amplifier **11** may be provided with multi-stage FETs **12** and **13**, respectively. The FETs **12** and **13** have drain source capacities Cds**1** and Cds**2**, respectively. The drain source capacitances Cds**1** and Cds**2** are internal capacitances of the FETs **12** and **13**, but are described by using dotted lines in FIG. **1**. The synthesizer **14** includes a drain source capacitance Cds**1** of the main amplifier **10**, an inductor L**0**, and a synthesis point N**1**. The drain source capacitance Cds**1** and the inductor L**0** function as an impedance inverter at a center frequency in a band of the Doherty amplifier and are used instead of the quarter wavelength line. At the synthesis point N**1**, the signal amplified by the main amplifier **10** and the signal amplified by the peak amplifier **11** are synthesized.

(31) The main amplifier **10** operates in class AB or class B, and the peak amplifier **11** operates in class C. Thereby, when an input power is small, the main amplifier **10** mainly amplifies the input signal. When the input power becomes large, the main amplifier **10** and the peak amplifier **11** amplify the input signal. Matching circuits may be connected between the distributor **16** and the main amplifier **10**, between the distributor **16** and the peak amplifier **11**, and between the synthesizer **14** and the output terminal Tout. When the peak amplifier **11** operates in a state where the input power of the high frequency signal input to the input terminal Tin is large, the main amplifier **10** and the peak amplifier **11** are set to operate optimally at a saturation power (for example, the efficiency is maximized). When the peak amplifier **11** does not operate in a state where the input power is small, the main amplifier **10** is set to operate optimally (for example, efficiency is maximized) at the saturation power.

(32) FIG. **2** is a circuit diagram of the Doherty amplifier according to the first embodiment. As illustrated in FIG. **2**, in the FET **12**, a plurality of FETs **12***a* to **12***d* are connected in parallel, and in the FET **13**, a plurality of FETs **13***a* to **13***d* are connected in parallel. The FETs **12** and **13** are multi-finger FETs, and the FETs **12***a* to **12***d* and the FETs **13***a* to **13***d* are schematically representations of multi-finger FETs. The FETs **12***a* to **12***d* and the FETs **13***a* to **13***d* have drain source capacities Cds**1***a* to Cds**1***d* and Cds**2***a* to Cds**2***d*, respectively.

(33) An inductor L**1** and a capacitor C**1** (first capacitor) are connected in series between the drains of the FETs **12***a* and **12***b* and the drains of the FETs **13***c* and **13***d*. An inductor L**2** and a capacitor C**2** (second capacitor) are connected in parallel between the drains of the FETs **12***c* and **12***d* and the drains of the FETs **13***a* and **13***b*. The drain source capacitances Cds**1***a* and Cds**1***b* of the FETs **12***a* and **12***b*, the inductor L**1** and the capacitor C**1** form a synthesizer **14***a*. The drain source capacitances Cds**1***c* and Cds**1***d* of the FETs **12***c* and **12***d*, the inductor L**2** and the capacitor C**2** form a synthesizer **14***b*.

(34) FIG. **3** is a plan view of the Doherty amplifier according to the first embodiment. As illustrated in FIG. **3**, the FETs **12** and **13** are provided on a semiconductor chip **40** and are arranged in a Y direction. The FET **12** is a multi-finger FET having a plurality of source fingers **23**, a plurality of gate fingers **24**, and a plurality of drain fingers **25**. The source fingers **23**, the gate fingers **24** and the drain fingers **25** extend in an X direction. The plurality of drain fingers **25** are connected to a drain electrode **20**. The plurality of gate fingers **24** are connected to a gate electrode **22**. The source fingers **23** are connected to a ground on a lower surface of the semiconductor chip **40** by via holes. The FET **13** includes a plurality of source fingers **33**, a plurality of gate fingers **34**, a plurality of drain fingers **35**, a drain electrode **30**, and a gate electrode **32**. The configurations of the source fingers **33**, the gate fingers **34**, the drain fingers **35**, the drain electrode **30**, and the gate electrode **32** in the FET **13** are the same as those in the FET **12**, and the description thereof will be omitted. Since the source fingers **23** and **33**, the gate fingers **24** and **34**, and the drain fingers **25** and **35** are arranged in the Y direction, the drain electrodes **20** and **30** and the gate electrodes **22** and **32** are widened in the Y direction.

(35) In the drain electrode **20** of the FET **12**, a region far from the drain electrode **30** of the FET **13**

is a first region **20***a*, and a region close to the drain electrode **30** is a third region **20***b*. In the drain electrode **30** of the FET **13**, a region far from the drain electrode **20** of the FET **12** is a second region **30***a*, and a region close to the drain electrode **20** is a fourth region **30***b*.

(36) A pad **26** connected to the region **20***a* is provided on a +X side of the region **20***a*. A pad **36** connected to the region **30***a* is provided on the +X side of the region **30***a*. A pad **38** is provided in a +Y side of the pad **36**. The capacitor C**1** is connected between the pads **36** and **38**. The pads **26** and **38** are connected by bonding wires **41**. Regions where the bonding wires **41** bond to the pads **26** and **38** are regions **20***c* and **30***c*, respectively. The regions **20***a* and **30***a* are connected to each other via the pad **26**, the bonding wires **41**, the pad **38**, the capacitor C**1** and the pad **36**. Distances D**4** (electrical length) between the region **20***c* and the plurality of drain fingers **25** connected to the region **20***a* are substantially the same as each other. Distances D**5** between the region **30***c* and the plurality of drain fingers **35** connected to the region **30***a* are substantially the same as each other.

(37) The regions **20***b* and **30***b* are connected by bonding wires **42**. Regions where the bonding wires **42** bond to the drain electrodes **20** and **30** are regions **20***d* and **30***d*, respectively. The capacitor C**2** is connected in parallel with the bonding wires **42** between the regions **20***d* and **30***d*. Distances D**3** between the region **20***d* and the plurality of drain fingers **25** connected to the region **20***b* are substantially the same as each other. Another distances D**3** between the region **30***d* and the plurality of drain fingers **35** connected to the region **30***b* are substantially the same as each other.

(38) Each of the capacitors C**1** and C**2** is a MIM (Metal Insulator Metal) capacitor having an upper electrode **44**, a lower electrode **46**, and a dielectric film **45**. The drain electrode **30** is connected to a matching circuit **18** via bonding wires **43**. A region where the bonding wires **43** bond to the drain electrode **30** is a region **30***e*. The matching circuit **18** is connected to the output terminal Tout. The drain electrodes **20**, **30**, the gate electrodes **22**, **32**, the pads **26**, **36** and **38**, the upper electrode **44**, and the lower electrode **46** are metal layers such as a gold layer. Bonding wires are connected between the distributor **16** and the gate electrode **22** and between the quarter wavelength line **15** and the gate electrode **32**, but the illustration of the bonding wires is omitted.

(39) FIG. **4** is a cross-sectional view of the capacitor in the first embodiment. As illustrated in FIG. **4**, the drain electrode **20** and the pad **38** (and the drain electrode **30**) are provided on the semiconductor chip **40**. The drain electrode **20** is connected to the upper electrode **44**, and the pad **38** (and the drain electrode **30**) is connected to the lower electrode **46**. In the capacitor C**1** (and C**2**), the dielectric film **45** is provided between the upper electrode **44** and the lower electrode **46**. The dielectric film **45** is an inorganic dielectric film such as a silicon nitride film or a silicon oxide film. An insulating film **45***a* is provided so as to cover the upper electrode **44**. Both ends of the bonding wire **41** (and **42**) are bonded to the regions **20***c* (and **20***d*) and the regions **30***c* (and **30***d*). The drain electrode **20** may be connected to the lower electrode **46** and the pad **38** (and the drain electrode **30**) may be connected to the upper electrode **44**.

First Comparative Example

(40) FIG. **5** is a plan view of a Doherty amplifier in a first comparative example. As illustrated in FIG. **5**, in the first comparative example, the capacitors C**1** and C**2** are not provided. The regions **20***c* and **30***c* to which the bonding wires **41** are bonded are between the regions **20***a* and **20***b* and between the regions **30***a* and **30***b*, respectively. The bonding wires **42** are not provided. Other configurations are the same as those in the first embodiment.

(41) In the first comparative example, the phases of the signals are almost the same at points where the plurality of drain fingers **25** (and **35**) are connected to the drain electrodes **20** (and **30**). The distance D**1** between the regions **20***c* (and **30***c*) and the drain fingers **25** (and **35**) far from the regions **20***c* (and **30***c*) is longer than the distance D**2** between the regions **20***c* (and **30***c*) and the drain fingers **25** (and **35**) close to the regions **20***c* (and **30***c*). Thereby, the phases of the signals output from the drain fingers **25** (and **35**) to be synthesized in the regions **20***c* (and **30***c*) are different from each other. Therefore, the loss increases.

Second Comparative Example

(42) FIG. **6** is a plan view of a Doherty amplifier in a second comparative example. As illustrated in FIG. **6**, the bonding wires **41** and **42** are provided. The regions **20***c* and **30***c* to which the bonding wires **41** are bonded are provided in the regions **20***a* and **30***b*, respectively. The regions **20***d* and **30***d* to which the bonding wires **42** are bonded are provided in the regions **20***b* and **30***b*, respectively. A reason why the bonding wires **41** connect the regions **20***c* and **30***c* and the bonding wires **42** connect the regions **20***d* and **30***d* is to make the lengths of the bonding wires **41** and **42** the same and make the respective inductances the same, which prevents the phases of the signals flowing through the bonding wires **41** and **42** from deviating from each other. Other configurations are the same as those in the first comparative example.
(43) In the second comparative example, the regions **20***c* and **20***d* (or **30***a* and **30***b*) are provided near the centers of the regions **20***a* and **20***b* (or **30***c* and **30***d*). Therefore, the distances D**3** between the drain finger **25** (or **35**) and the regions **20***c* and **20***d* (or **30***c* and **30***d*) are almost the same as each other. Therefore, the phases of the signals output from the drain fingers **25** (or **35**) to be synthesized in the regions **20***c* and **20***d* (or **30***c* and **30***d*) are almost the same as each other. Thereby, the loss can be suppressed. Although the example in which the regions **20***a* and **20***b* (or **30***a* and **30***b*) are provided for two drain fingers **25** (or **35**) is described, the regions **20***a* and **20***b* (or **30***a* and **30***b*) may be provided for three drain fingers **25** (or **35**). Even in that case, the phases of the signals synthesized in the regions **20***c* and **20***d* (or **30***c* and **30***d*) can be aligned in the second comparative example, as compared with the first comparative example. However, in the second comparative example, in order to align the lengths of the bonding wires **41** and **42** to be the same, the bonding wires **41** and **42** are connected so as to intersect each other. However, due to the proximity effect, the resistance values of the bonding wires **41** and **42** increase, and the Q values of the bonding wires **41** and **42** (that is, the inductors L**1** and L**2**) decrease.
Third Comparative Example
(44) FIG. **7** is a circuit diagram of a Doherty amplifier in a third comparative example. FIG. **8** is a plan view of the Doherty amplifier in the third comparative example. As illustrated in FIGS. **7** and **8**, in the third comparative example, the regions **20***c* and **30***c* to which the bonding wires **41** are bonded are provided in the regions **20***a* and **30***a*, respectively. The regions **20***d* and **30***d* to which the bonding wires **42** are bonded are provided in the regions **20***b* and **30***b*, respectively. The bonding wires **41** and **42** correspond to the inductors L**1** and L**2**, respectively. Other configurations are the same as those in the second comparative example.
(45) The distances D**3** between the drain finger **25** and the regions **20***c* and **20***d* are almost the same as each other. Thereby, the phases of the signals in the regions **20***c* and **20***d* are almost aligned. Similarly, the distances D**3** between the drain finger **35** and the regions **30***c* and **30***d* are almost the same each other, and the phases of the signals in regions **30***c* and **30***d* are almost aligned. However, the lengths of the bonding wires **41** and **42** are different from each other, and the inductances of inductors L**1** and L**2** are different from each other. Further, the electric length from the region **20***a* to the region **30***e* via the bonding wires **41** and the region **30***a* are different from the electric length from the region **20***b* to the region **30***e* via the bonding wires **42** and the region **30***b*. This makes it difficult to set both of the synthesizers **14***a* and **14***b* to have an electrical length of the quarter wavelength at the center frequency of the band. Even if the lengths of the bonding wires **41** and **42** are about the same as each other, the electrical length between the region **20***a* and the synthesis point N**1** and the electrical length between the region **20***b* and the synthesis point N**1** are different from each other.
(46) According to the first embodiment, as illustrated in FIG. **3**, the bonding wires **41** (first bonding wire) are connected between the region **20***a* (first region) in the drain electrode **20** (first output electrode) and the region **30***a* (second region) in the drain electrode **30** (second output electrode). The bonding wires **42** (second bonding wire) are connected between the region **20***b* (third region) different from the region **20***a* in the drain electrode **20** and the region **30***b* (fourth region) different from the region **30***a* in the drain electrode **30**. In this way, when the different regions **20***a* and **20***b*

in the drain electrode **20** are connected to the different regions **30***a* and **30***b* in the drain electrode **30**, respectively, the electrical lengths of the synthesizers **14***a* and **14***b* are different from each other, and the characteristics of the Dougherty amplifier are deteriorated. Therefore, the capacitor C**1** (first capacitor) connected in series or in parallel with the bonding wires **41** is provided between the regions **20***a* and **30***a*. The capacitor C**2** (second capacitor) connected in series or in parallel with the bonding wires **42** is provided between the regions **20***b* and **30***b*. Thereby, the electric lengths of the synthesizers **14***a* and **14***b* can be made substantially the same. Therefore, the deterioration of the characteristics of the Doherty amplifier can be suppressed.
(47) The region **20***b* is closer to the drain electrode **30** than the region **20***a*, and the region **30***b* is closer to the drain electrode **20** than the region **30***a*. At this time, the distance between the region **20***a* and the synthesis point N**1** is longer than the distance between the region **20***b* and the synthesis point N**1**. For this reason, the capacitor C**1** is connected in series with the bonding wires **41** between the regions **20***a* and the regions **30***a*. The capacitor C**2** is connected in parallel with the bonding wires **42** between the regions **20***b* and the regions **30***b*. Thereby, a total electrical length of the connection between the region **20***a* and the synthesis point N**1** via the series connection of the bonding wires **41** and the capacitor C**1** can be substantially shorter than the total electrical length in the case where the capacitor C**1** is not included between the region **20***a* and the synthesis point N**1**. The total electrical length of the connection between the region **20***b* and the synthesis point N**1** via the parallel connection of the bonding wires **42** and capacitor C**2** can be substantially longer than the total electrical length in the case where the capacitor C**2** is not included. Therefore, the electrical length between the region **20***a* and the synthesis point N**1** and the electrical length between the region **20***b* and the synthesis point N**1** can be almost the same as each other.
(48) The main amplifier **10** includes the plurality of drain fingers **25** (first output fingers), and the regions **20***a* and **20***b* are regions to which the plurality of drain fingers **25** are connected. Further, the peak amplifier **11** includes the plurality of drain fingers **35** (second output fingers), and the regions **30***a* and **30***b* are regions to which the plurality of drain fingers **35** are connected. In such multi-finger FETs, the phases are substantially the same as each other in the regions **20***a* and **20***b* to which the drain fingers **25** are connected. Therefore, when the electrical length between the region **20***a* and the synthesis point N**1** is different from the electrical length between the region **20***b* and the synthesis point N**1**, the characteristics of the Doherty amplifier deteriorate. Therefore, it is preferable to provide capacitors C**1** and C**2**.
(49) In the multi-finger FET, the widths of the drain electrodes **20** and **30** in a direction intersecting the extension direction (X direction) of the drain fingers **25** and **35** are widened. Therefore, when the main amplifier **10** and the peak amplifier **11** are arranged in the direction intersecting the extension direction of the drain fingers **25** and the drain fingers **35**, respectively, the electrical length between the region **20***a* and the synthesis point N**1** and the electrical length between the region **20***b* and the synthesis point N**1** easily differ from each other. Therefore, it is preferable to provide capacitors C**1** and C**2**.
(50) The regions **20***d* and **30***d* to which the bonding wires **42** are bonded are provided in the regions **20***b* and **30***b*, respectively. On the other hand, the pads **26** and **36** for pulling out the regions **20***a* and **30***a* in the +X direction are provided, respectively, and the bonding wires **41** are bonded in the regions **20***c* and **30***c* which are arranged in the +X direction from the regions **20***a* and **30***a*. Thereby, it is possible to prevent the bonding wires **41** and **42** from intersecting with each other. Therefore, it is possible to suppress a decrease in the Q value as illustrated in the second comparative example.
(51) Further, the signal output by the main amplifier **10** and the signal output by the peak amplifier **11** are synthesized in the drain electrode **30**. Thereby, a difference easily occurs between the electric length between the region **20***a* and the synthesis point N**1** and the electric length between the region **20***b* and the synthesis point N**1**. Therefore, it is preferable to provide the capacitors C**1** and C**2**.
(52) When the regions **20***a* and **30***a* are connected to each other via the region **20***b*, the signal from

the drain finger **25** connected to the region **20***a* and the signal from the drain finger **25** connected to the region **20***b* are synthesized in the region **20***b*. Thereby, the signals having different phases are synthesized and the loss occurs. Similarly, when the regions **20***a* and **30***a* are connected to each other via the region **30***b*, the signals having different phases are synthesized in the region **30***b* and the loss occurs. According to the first embodiment, the regions **20***a* and **30***a* are connected to each other without through the regions **20***b* and **30***b*, and the regions **20***b* and **30***b* are connected to each other without through the regions **20***a* and **30***a*. Thereby, the loss can be suppressed.

First Variation of First Embodiment

(53) FIG. **9** is a plan view of a Doherty amplifier in a first variation of the first embodiment. As illustrated in FIG. **9**, the pad **36** is pulled out from the region **30***a* in the +X direction and further bent in the +Y direction. The region **30***c* to which the bonding wire **41** is bonded is provided on the pad **36**. The capacitor C**1** is not connected in series or in parallel with the bonding wire **41** between the regions **20***a* and **30***a*. The capacitor C**2** is connected in parallel with the bonding wires **42** between the regions **20***b* and **30***b*. Other configurations are the same as those in the first embodiment, and the description thereof will be omitted. Only the capacitor C**2** among the capacitors C**1** and C**2** may be provided as in the first variation of the first embodiment.

Second Variation of First Embodiment

(54) FIG. **10** is a plan view of a Doherty amplifier in a second variation of the first embodiment. As illustrated in FIG. **10**, the capacitor C**1** is connected in series with the bonding wires **41** between the regions **20***a* and **30***a*. The capacitor C**2** is not connected in series or in parallel with the bonding wires **42** between the regions **20***b* and **30***b*. Other configurations are the same as those in the first embodiment, and the description thereof will be omitted. Only the capacitor C**1** among the capacitors C**1** and C**2** may be provided as in the second variation of the first embodiment.

(55) At least one of the capacitors C**1** and C**2** may be provided as in the first embodiment and the first and second variations thereof. Whether both of the capacitors C**1** and C**2** are provided or any one of the capacitors C**1** and C**2** are provided can be appropriately designed so that the electrical length between the region **20***a* and the synthesis point N**1** is almost the same as the electrical length between the region **20***b* and the synthesis point N**1**.

(56) Although the example in which the FETs **12** and **13** are provided on the same semiconductor chip **40** is described, the FETs **12** and **13** may be provided on different semiconductor chips. Although the example in which the main amplifier **10** and the peak amplifier **11** include the FETs **12** and **13**, respectively, is described, the main amplifier **10** and the peak amplifier **11** may include transistors other than the FETs. Although the example of using the MIM capacitors as the capacitors C**1** and C**2** is described, capacitors other than the MIM capacitors may be used.

(57) An example in which the drain electrode **20** is set to the two regions **20***a* and **20***b* and the drain electrode **30** is set to the two regions **30***a* and **30***b* is described. However, the drain electrode **20** may be set to three or more regions and the drain electrode **30** may be set to three or more regions. For example, the region between the regions **20***a* and **20***b* in the drain electrode **20** and the region between the regions **30***a* and **30***b* in the drain electrode **30** may be connected by bonding wires other than the bonding wires **41** and **42**.

(58) Although the examples of the main amplifier **10** and the peak amplifier **11** are described as the first amplifier and the second amplifier, respectively, the first amplifier and the second amplifier may be the peak amplifier **11** and the main amplifier **10**, respectively.

Second Embodiment

(59) FIG. **11** is a plan view of a Doherty amplifier in a second embodiment.

(60) As illustrated in FIG. **11**, the regions **20***a* and **20***b* in the drain electrode **20** are divided by a division region **29**. The regions **30***a* and **30***b* in the drain electrode **30** are divided by a division region **39**. The region **30***b* and the pad **36** are electrically connected to each other via the lower electrode **46** of the capacitor C**1** and without via the region **30***a*. Other configurations are the same as those in the first embodiment, and the description thereof will be omitted.

First Variation of Second Embodiment

(61) FIG. **12** is a plan view of a Doherty amplifier in a first variation of the second embodiment. As illustrated in FIG. **12**, the capacitor C**1** is not connected in series or in parallel with the bonding wires **41** between the regions **20***a* and **30***a*. The division regions **29** and **39** in the drain electrodes **20** and **30** are provided. The regions **30***a* and **30***b* are not directly connected to each other, but are connected to each other via the pad **36**. The capacitor C**2** is connected in parallel with the bonding wires **42** between the regions **20***b* and **30***b*. Other configurations are the same as those in the second embodiment, and the description thereof will be omitted. Only the capacitor C**2** among the capacitors C**1** and C**2** may be provided as in the first variation of the second embodiment.

Second Variation of Second Embodiment

(62) FIG. **13** is a plan view of a Doherty amplifier in a second variation of the second embodiment. As illustrated in FIG. **13**, the capacitor C**1** is connected in series with the bonding wires **41** between the regions **20***a* and **30***a*. The capacitor C**2** is not connected in series or in parallel with the bonding wires **42** between the regions **20***b* and **30***b*. Other configurations are the same as those in the second embodiment, and the description thereof will be omitted. Only the capacitor C**1** among the capacitors C**1** and C**2** may be provided as in the second variation of the second embodiment.

(63) When the regions **20***a* and **20***b* are connected and the regions **30***a* and **30***b* are connected as in the first embodiment and the variations thereof, the characteristics are improved, but oscillation is likely to occur. Therefore, the regions **20***a* and **20***b* are separated and the regions **30***a* and **30***b* are separated as in the second embodiment and the variations thereof. Thereby, the oscillation can be suppressed.

Third Embodiment

(64) FIG. **14** is a circuit diagram of a Doherty amplifier in a third embodiment. As illustrated in FIG. **14**, a harmonic processing circuit **48** is connected between the synthesis point N**1** and the ground. The harmonic processing circuit **48** includes an inductor L**3** and a capacitor C**3** that are connected in series between the synthesis point N**1** and the ground. The inductor L**3** and the capacitor C**3** form a series resonant circuit and have a resonant frequency in a harmonic frequency band in a band of the Doherty amplifier. Other circuit configurations are the same as those in FIG. **2** of the second embodiment, and the description thereof will be omitted.

(65) FIG. **15** is a plan view of the Doherty amplifier in the third embodiment. As illustrated in FIG. **15**, the capacitor C**3** includes a dielectric film **49**, an upper electrode **51**, and a lower electrode (not illustrated). A bonding wire **47** corresponds to the inductor L**3** and connects the drain electrode **30** and the upper electrode **51**. The lower electrode is connected to the ground. Other configurations are the same as those in the first embodiment, and the description thereof will be omitted.

(66) According to the third embodiment, the harmonic processing circuit **48** is connected to the drain electrode **30** and processes the harmonic components of the signals amplified by the main amplifier **10** and the peak amplifier **11**. By providing the harmonic processing circuit **48** in the drain electrode **30** corresponding to the synthesis point N**1**, the single harmonic processing circuit **48** can be used to process the harmonics of both the main amplifier **10** and the peak amplifier **11**. Therefore, the Doherty amplifier can be reduced in size. The harmonic processing circuit **48** may be provided in the variations of the first embodiment, the second embodiment and the variations of the second embodiment.

(67) The embodiments disclosed here should be considered illustrative in all respects and not restrictive. The present disclosure is not limited to the specific embodiments described above, but various variations and changes are possible within the scope of the gist of the present disclosure as described in the claims.

Claims

1. A Doherty amplifier comprising: a first amplifier that includes a plurality of first output fingers and a first output electrode connected to the plurality of first output fingers, amplifies one of two signals into which an input signal is distributed, and outputs an amplified signal to the first output electrode; a second amplifier that includes a plurality of second output fingers and a second output electrode connected to the plurality of second output fingers, amplifies another one of the two signals, and outputs an amplified signal to the second output electrode; a first bonding wire connected between a first region in the first output electrode and a second region in the second output electrode; a second bonding wire connected between a third region in the first output electrode closer to the second output electrode than the first region and a fourth region in the second output electrode closer to the first output electrode than the second region; and at least one of a first capacitor connected in series with the first bonding wire between the first region and the second region, and a second capacitor connected in parallel with the second bonding wire between the third region and the fourth region; wherein the first region and the third region are regions to which the plurality of first output fingers are connected, the second region and the fourth region are regions to which the plurality of second output fingers are connected, and the first amplifier and the second amplifier are arranged in a direction intersecting extension directions of the first output fingers and the second output fingers, respectively.
2. The Doherty amplifier according to claim 1, wherein the first output electrode is separated between the first region and the third region, and the second output electrode is separated between the second region and the fourth region.
3. The Doherty amplifier according to claim 1, wherein the first bonding wire does not intersect with the second bonding wire.
4. The Doherty amplifier according to claim 1, wherein the first region and the second region are connected to each other without through the third region and the fourth region, and the third region and the fourth region are connected to each other without through the first region and the second region.
5. The Doherty amplifier according to claim 1, wherein a signal output by the first amplifier and a signal output by the second amplifier are synthesized in the second output electrode.
6. The Doherty amplifier according to claim 5, further comprising: a harmonic processing circuit that is connected to the second output electrode and processes a harmonic component of the signals amplified by the first amplifier and the second amplifier.
7. The Doherty amplifier according to claim 1, wherein a capacitor included in the Doherty amplifier is the first capacitor.
8. The Doherty amplifier according to claim 1, wherein a capacitor included in the Doherty amplifier is the second capacitor.
9. The Doherty amplifier according to claim 1, wherein capacitors included in the Doherty amplifier are the first capacitor and the second capacitor.